friction; hence the machines are protected from wear and tear over the years.

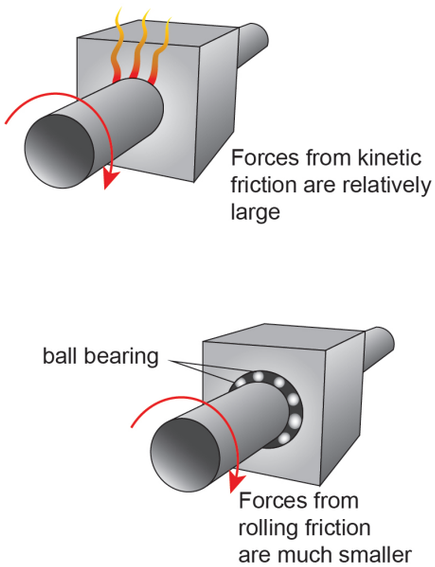


Figure 3.35 Reducing kinetic friction using ball bearing

During the time of Newton and Galileo, frictional force was considered as one of the natural forces like gravitational force. But in the twentieth century, the understanding on atoms, electron and protons has changed the perspective. The frictional force is actually the electromagnetic force between the atoms on the two surfaces. Even well polished surfaces have irregularities on the surface at the microscopic level as seen in the Figure 3.36.

POINTS TO PONDER

When you walk on the tiled floor where water is spilled, you are likely to slip. Why?



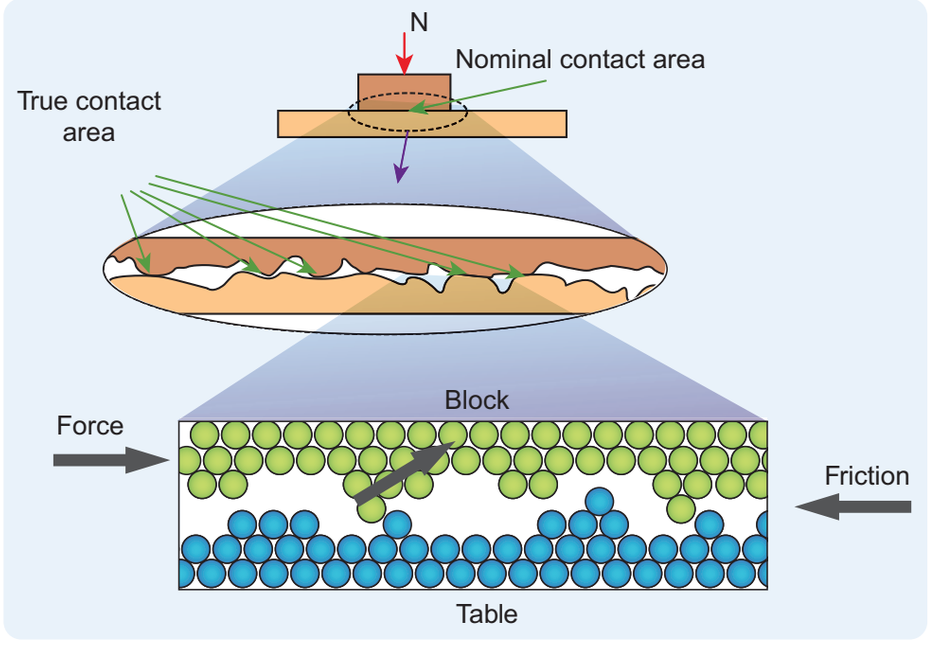


Figure 3.36 Irregularities on the surface at the microscopic level

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